

COMMISSIONING DATA REVIEW

Butane steam-generating heat pump

Corrected reconstruction of commissioning on 26 August 2026

Record	Value
Plant	Ahuora butane steam-generating heat pump
Recorded window	08:01–17:39 NZST, 26 August 2026
Data	InfluxDB lab/data; 53 PLC topics; 30 s, 5 s and 2 s working extracts
Reference drawing	P&ID V22, Rev 22, 27 July 2026
Control mode	Manual, except flash-tank level PID
Status	Working record incorporating operator corrections; remaining scaling questions are explicit

Revised headline Commissioning moved from wet morning bump/low-load tests to one dry startup and a deliberate manual ramp. Atmospheric steam was vented before STGV61 was closed for the final pressurisation. The principal unresolved process problem is liquid carryover, strongly coupled to a 9–10 minute source-side thermal cycle with one abnormal 18.25-minute interval.

1. Corrections that change the interpretation

Signal / topic	Correct interpretation	Consequence
HPEV01	Recorded percent closed: 95% means about 5% open	Previous “95% open/saturated” interpretation is reversed.
HPCO01	Speed command above the 30 Hz minimum; 0% can mean stopped or running at 30 Hz	Never use the zero transition alone as a start/stop contact. Infer operation from pressures, temperatures, EEV action and pump-down evidence.
HPEV01_Superheat	Uses evaporator-inlet pressure HPPT06 and is not compressor-inlet superheat	Do not use for judging liquid entering the compressor; identify its paired temperature before treating it as evaporator superheat.
HPPT06	Pressure after the expansion valve and before the evaporator, kPa gauge	Compare with HPPT01 during starts and pump-down. The configured shutdown threshold was 20 kPa gauge.
HPPT01 + HPPT01	Compressor-suction temperature and pressure	Use this pair for suction superheat; P&ID locates both immediately upstream of the compressor.
STPT61 data	Steam temperature; raw unit 0.1 °C	This is the temperature record despite the PT-style tag name: 1006 = 100.6 °C; 1253 = 125.3 °C.
All pressure tags	Engineering unit is kPa	Gauge/absolute reference must still be recorded per transmitter.
STTT61 data	Steam pressure signal, kPa	This is the pressure record despite the TT-style tag name; it remains near 63–64 kPa during known pressurisation and is not presently usable.
STGV61O	Steam-valve command/output	There is no actual valve-position feedback. Exclude presumed feedback from analysis.
HPGD01 startup	Expected detector startup behaviour after extended power-off	Document as a startup response, not evidence of a leak, unless other evidence appears.
Plant control	Manual actions except flash-tank level PID	Compressor steps are deliberate operator actions, not automatic pressure control. Actual compressor start/stop state was not logged.

2. What happened

Time (NZST)	Phase	Evidence / interpretation
08:01–09:09	Power-up and circulation checks	Telemetry starts; detector startup response begins near 08:28. Common zeros likely reflect power/program/logger transitions.
Morning	Compressor bump tests	Short checks followed by an inferred 09:30–11:35 low-load operating interval. Operators observed liquid passing through the compressor.
09:30–11:35	Extended low-load operation and pump-down	Compressor command near 24%, then 0%/30 Hz during pump-down. HPPT06 crosses 20 kPa at 11:35:20 and the EEV record changes to 0% shortly afterward; physical stop is inferred, not directly recorded.
12:07–13:51	Repeated manual tests	Several higher-load tests. Command-zero periods around 12:22, 13:15 and 13:37 are not necessarily stops; pressure evidence indicates some are continued 30 Hz operation or pump-down.
13:55–14:03	High-pressure cutoff calibration	A restart probably occurs during the telemetry gap while command is still 0%/30 Hz. Command is ramped 24 → 48 → 60 → 72%; HPPT02 peaks at 2765 kPa at 14:03:04 and collapses while the command remains 72%, identifying the hardware cutoff.
14:06–14:11	Post-trip pump-down	Process evidence indicates probable 30 Hz operation despite a 0% command. HPPT06 crosses 20 kPa at 14:10:20; the EEV record changes to 0% near 14:11 and the compressor is inferred stopped.
14:47–14:52	Restart permissives	Circulation re-established; EEV recorded near 95% closed (~5% open).
14:52–15:04	Final start and deliberate ramp	The compressor probably starts at 30 Hz near 14:51:50, about 2.5 min before command rises to 24%, then ramps 24 → 48 → 72 → 96%.
15:05–16:10	Atmospheric steam generation	STPT61 stabilises at 100.65 °C; operator reports steam pressure near 0 barg.
16:12–16:26	Final pressurised test	STGV61 closure begins at 16:12:15 and reaches 50/1000 command at 16:16. Operator reports ~1 barg; steam temperature peaks at 125.3 °C.
16:33–16:53	Unload and cooldown	Command is staged down; process response indicates the compressor probably stops near 16:40. Circulation continues for about 12 minutes.

Butane steam-generating heat pump commissioning overview



Figure 1. Corrected day overview. EEV is shown as inferred opening (100 - recorded % closed); STPT61 is plotted as steam temperature.

3. Corrected compressor-suction analysis

The P&ID places HPPT01 and HPPT01 together on the suction line, downstream of the suction accumulator and directly before the compressor. Superheat was recalculated at 2 s resolution using n-butane saturation properties, HPPT01/10 in °C, and HPPT01 provisionally interpreted as kPa gauge with 101.325 kPa added to obtain absolute pressure.

Interpret carefully A result at or below 0 K means the measured suction state calculates at/below saturation. That is consistent with the team's observation of liquid carryover, but the negative magnitude should not be treated literally until pressure units/reference, calibration, time alignment and sensor uncertainty are confirmed.

Operating-state disclaimer The actual compressor start/stop contact was not recorded. HPCO01=0% may mean stopped or running at the 30 Hz minimum. The intervals below are revised estimates using command changes together with HPPT02/HPPT02 response, HPPT06 pump-down, HPPT01, EEV action and pressure equalisation. Their boundaries are not verified motor-state timestamps.

Estimated operating interval	Confidence	Duration	Median SH	At/below 0 K	Interpretation
09:30:30–11:35:32	High	124.8 min	–2.42 K	98.9%	Wet/at saturation almost throughout; includes inferred 30 Hz pump-down.
12:07:20–12:29:20	Medium	22.0 min	–1.97 K	92.1%	Probable 30 Hz start and continued pump-down after command returns to zero.
12:37:46–12:48:45	High	11.0 min	–2.29 K	100.0%	Entire interval calculates at/below saturation.
12:55:52–13:51:10	Medium	55.3 min	–1.78 K	96.8%	13:15, 13:22, 13:37 and 13:49 are probably speed changes/30 Hz operation, not separate stop-start cycles.
~13:55:30–14:03:04	Medium	7.6 min	–1.28 K	78.5%	Restart occurs within/at the end of a telemetry gap; ends at the hardware high-pressure cutoff.
14:05:50–14:10:40	Medium	4.8 min	+11.29 K	41.1%	Probable 30 Hz restart and pump-down; first two minutes still calculate wet.
14:51:50–16:40:05	High	108.2 min	+11.48 K	15.6%	Only dry startup; still has distinct returns to saturation/liquid risk.

The revised inferred operating time is about 5.57 h, versus 4.89 h if only command-above-zero samples are counted. Across the inferred intervals, 69.7% of samples are at/below 0 K, 70.9% are ≤ 2 K and 72.5% are ≤ 5 K. During the final run, notable at/below-saturation intervals remain about 15:24:38–15:31:26, 16:18:12–16:21:08 and 16:28:10–16:35:18 NZST.

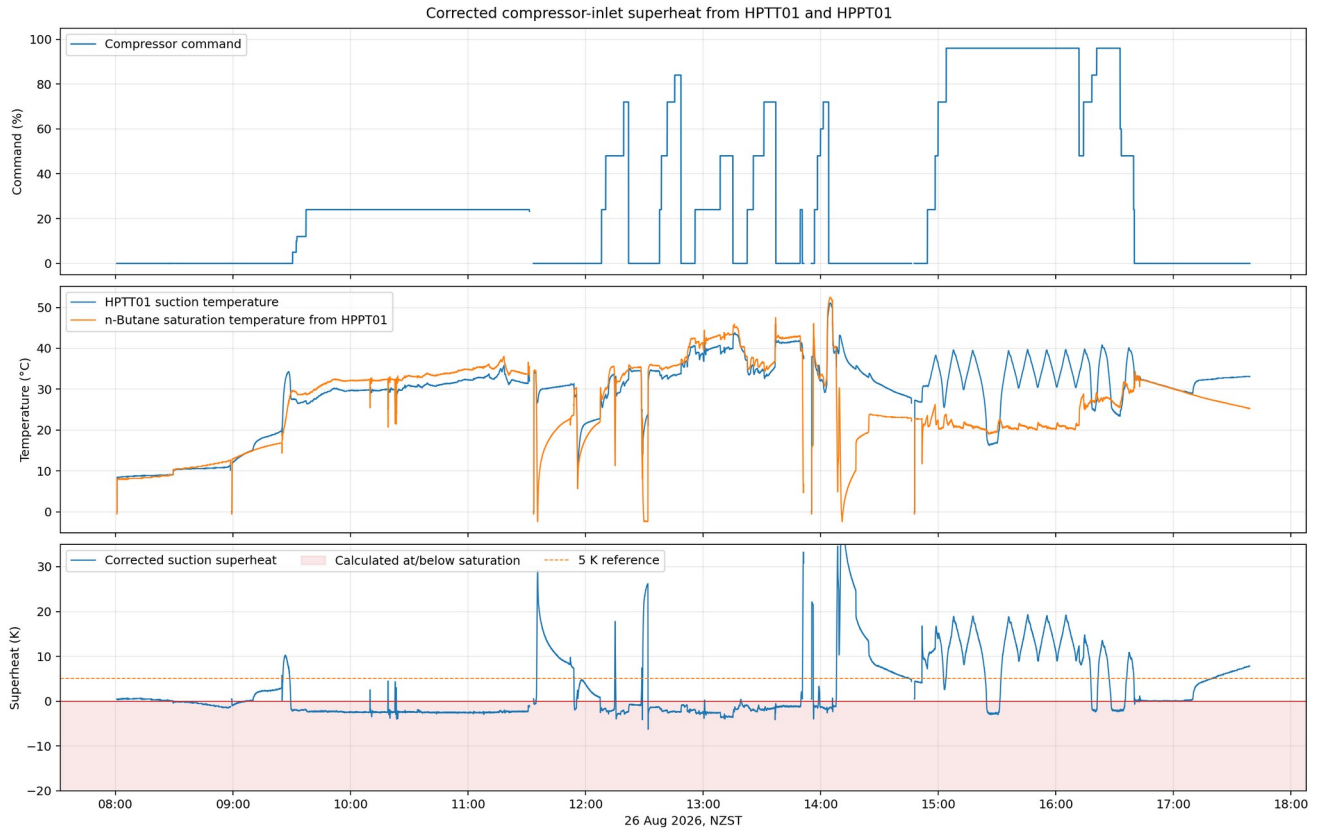


Figure 2. Recalculated compressor-suction superheat using HPTT01 and HPPT01. Thermodynamic conversion assumes n-butane and HPPT01 in kPa gauge.

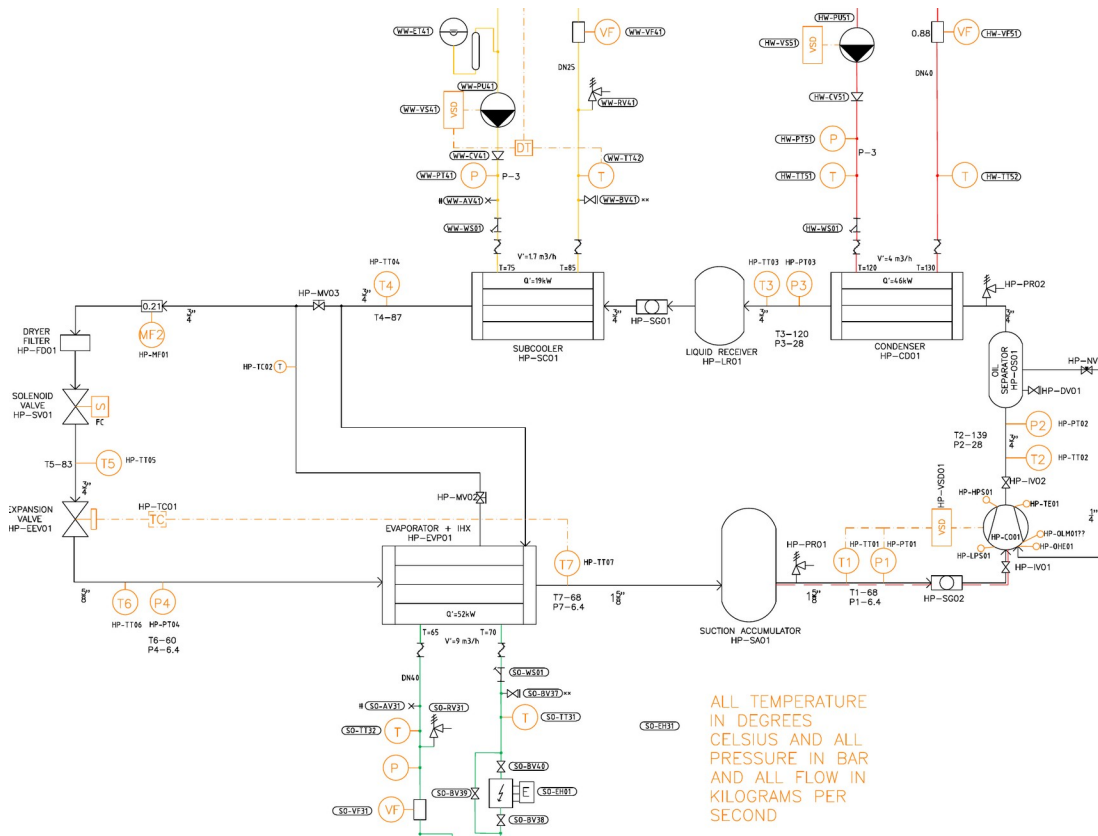


Figure 3. P&ID V22 excerpt: HPTT01 and HPPT01 are collocated on the compressor suction immediately after the suction accumulator.

3.1 High-side state and pressure ratio

A second thermodynamic pass used the P&ID interpretation that HPTT02/HPPT02 describe the hot-gas line upstream of the condenser, while HPTT03/HPPT03 describe the condenser outlet. As with the suction calculation, all pressure tags were provisionally treated as kPa gauge and converted to absolute pressure. These results are diagnostic indicators, not validated state points, until the transmitter reference and exact sensor locations are loop-checked.

Indicator	Atmospheric venting 15:08–16:10	Pressurised test 16:17–16:24	Interpretation
High-side saturation temperature	108.8 °C median	~123.9 °C median	Consistent with the deliberate rise in condensing pressure during STGV61 closure.
Absolute pressure ratio, P2/P1	8.49 median	8.84 median; 10.08 maximum	The compressor operates at a high ratio throughout and reaches its highest ratio during pressurisation.
Corrected suction superheat	13.55 K median	6.44 K median; 5th percentile –2.14 K	Suction dryness deteriorates during pressurisation and intermittently returns to calculated saturation.
HPTT02/HPPT02 superheat	11.70 K median	1.75 K median; minimum –0.94 K	Very little calculated hot-gas superheat during pressurisation; confirm the pair is a true, colocated discharge state.
HPPT03/HPTT03 subcooling	13.30 K median	9.98 K median	Positive calculated condenser-outlet subcooling, but it collapses during some liquid-risk intervals.

High-side caution A negative “discharge superheat” is not physically credible for a true compressor-discharge state. Plausible causes include a temperature sensor downstream of the stated location, sensor lag, a pressure/temperature location mismatch, wrong gauge/absolute basis or wet compression. The result strengthens the case for a loop check; it does not by itself prove liquid at the discharge sensor.

The source cycle is visible beyond the suction line. After detrending the vented-production period, SOTT31 correlates strongly with corrected suction superheat ($r=0.942$), and also with pressure ratio ($r=0.753$), calculated condenser-outlet subcooling ($r=0.697$), HPPT01 ($r=0.674$) and HPPT02 ($r=0.645$). The lag estimates are only about 1–3 minutes and should be treated as indicative because the signals are cyclic, but they show that the auxiliary source heat pump perturbs the whole refrigerant circuit rather than only the source-water temperature.

The 4 kg butane charge is a useful inventory fact, but this dataset cannot determine whether it is correct or excessive. That requires the design charge basis, receiver/accumulator volumes and levels, pipework inventory, operating conditions and a validated expansion-valve position. Persistent condenser subcooling together with wet suction while the EEV is mostly closed makes refrigerant distribution and stored liquid inventory a priority for the next test.



Figure 4. Final-run refrigerant-state indicators. Orange shading marks steam pressurisation. Calculations assume n-butane and all plotted pressure transmitters in kPa gauge; confirm before using absolute values.

4. Final run and steam production

From about 15:05, steam was vented to atmosphere. STPT61 therefore sits near 100 °C because water is boiling at atmospheric pressure; this is expected phase-equilibrium behaviour, not evidence of unusually stable control. No steam mass or condensate measurement is available. The refrigerant charge was approximately 4 kg of butane.

Final pressurisation is directly visible in the command data. STGV610 begins closing STGV61 at 16:12:15, moves from 1000 to 600, reaches 50/1000 at 16:16, and begins reopening at 16:24:55. Over this test, steam temperature rises from about 100.7 °C to 125.4 °C and HPPT02 rises from about 1705 to a peak 2579 kPa. The operator record places steam pressure at approximately 1 barg.

STTT61 pressure-channel issue STPT61 is the steam-temperature record. The separate STTT61 data are the steam-pressure record; although their engineering unit is kPa, they remain around 63–64 kPa during the known pressurisation and do not follow valve closure or the operator pressure observation. Do not use STTT61 to reconstruct pressure until loop-checked.

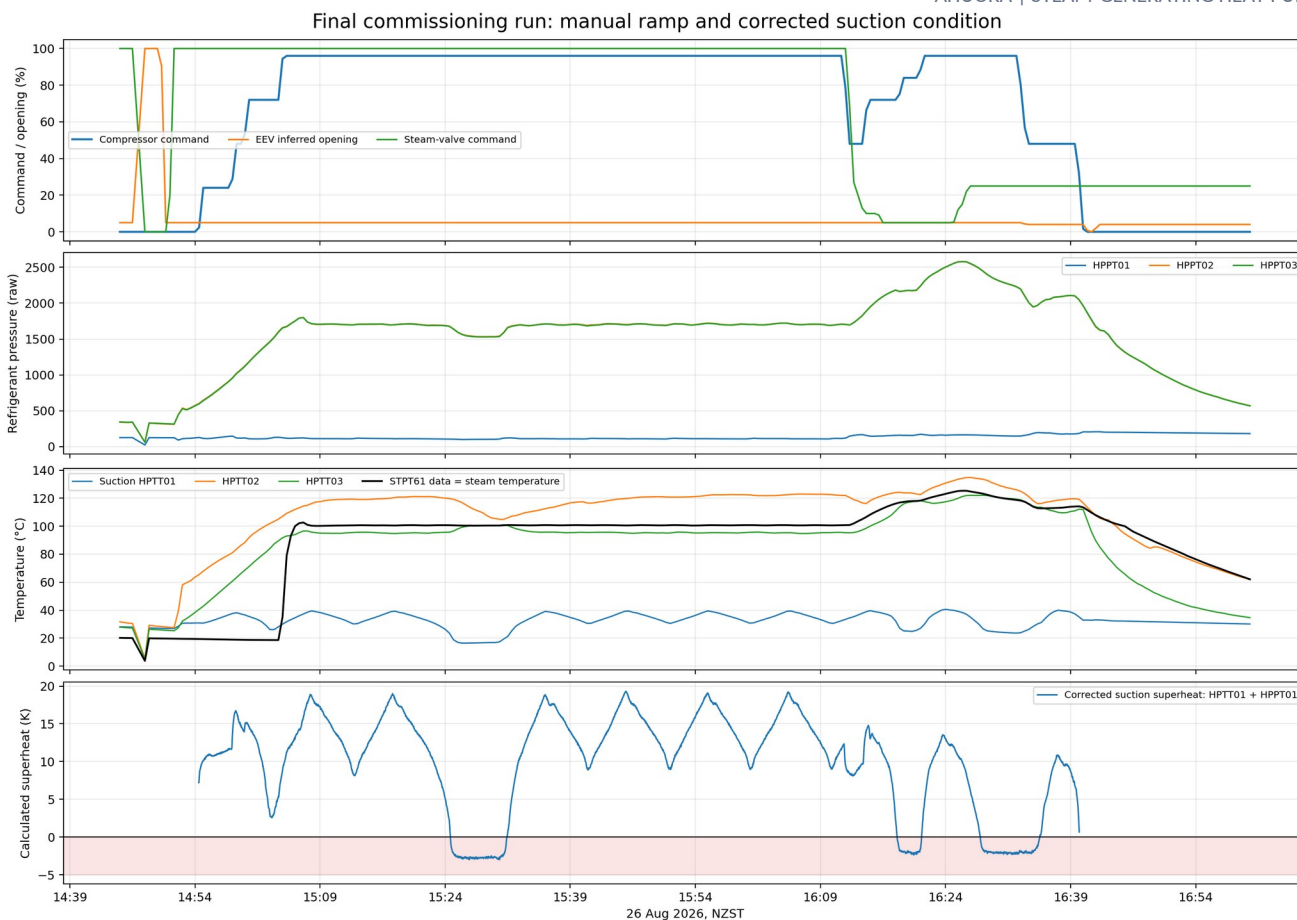


Figure 5. Final run with corrected signal meanings. The valve trace is command only; the lower panel uses the corrected suction-superheat calculation.

5. Source-loop and auxiliary heat-pump investigation

P&ID V22 shows the source loop, hold-up tank and external propane heat-pump exchanger around SOTT31, SOTT32 and SOTT33. The auxiliary heat pump has no logged run contact, so the blue phases below are inferred from SOTT31 changing from warming to cooling; they are not confirmed compressor states.

Inferred cooling start	Inferred stop	Next start	Run	Off	Start-to-start
14:58:20	15:02:30	15:07:25	4.17 min	4.92 min	9.08 min
15:07:25	15:12:30	15:17:15	5.08 min	4.75 min	9.83 min
15:17:15	15:26:10	15:35:30	8.92 min	9.33 min	18.25 min
15:35:30	15:40:30	15:45:15	5.00 min	4.75 min	9.75 min
15:45:15	15:50:15	15:55:00	5.00 min	4.75 min	9.75 min
15:55:00	15:59:55	16:04:40	4.92 min	4.75 min	9.67 min

Excluding the abnormal cycle, inferred cooling runs last 4.17–5.08 min (median 5.00), off periods 4.75–4.92 min (median 4.75), and start intervals 9.08–9.83 min. SOTT31 cools at about 2.27–2.78 K/min and warms at 2.34–2.48 K/min. The abnormal cycle cools for 8.92 min at 1.89 K/min, remains off/warming for 9.33 min at 1.89 K/min, and reaches 16.4 K total temperature drop.

Relationship during 15:00–16:10	Result	Meaning
Source circulation	SOVS31/SOVS32 fixed at 100%; flow median ~345 raw, SD 3.2	Temperature cycling is not a logged source-pump dropout.
SOTT31 ↔ HPPT01	Detrended $r=0.934$; suction response within about 1 min	Source temperature propagates strongly and quickly to compressor suction temperature.

Relationship during 15:00–16:10	Result	Meaning
Source ↔ suction superheat	Positive association; worst cold interval has SH median -2.73 K	Deep auxiliary cooling can drive the suction state to saturation/liquid risk.
SOTT31 ↔ WWTT41/WWTT43	Raw $r=0.99$; median differences only 0.12 – 0.16 K	These nominally separate tags behave almost like the same thermal inventory; check hydraulic arrangement and tag locations.
Source ↔ hot-water/steam temperatures	Only weak cycle correlation	The short source cycle is strongly attenuated at the steam-generating sink.

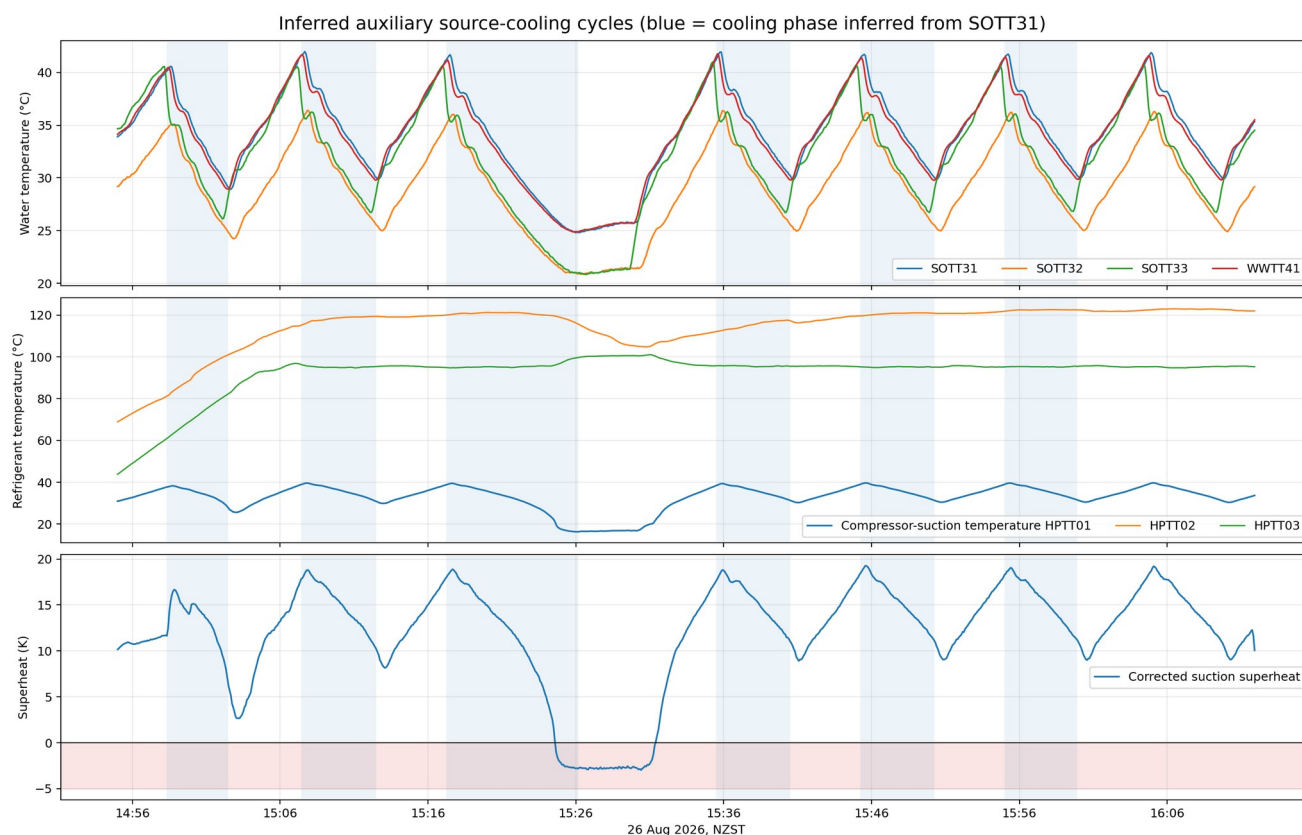


Figure 6. Detailed inferred source-cooling cycles. Blue shading is peak-to-valley cooling inferred from SOTT31, not a directly logged auxiliary-compressor state.

Source-loop conclusion The normal ~ 9.7 min cycle is consistent with roughly five minutes cooling and five minutes off. The 18.25 min cycle is the main candidate for the reported random auxiliary-heat-pump interruption. Direct run, demand and fault signals are required to decide whether it was a trip, timer lockout, thermostat cycle or manual restart.

6. Startup and shutdown procedure review

Startup quality is judged from the first two minutes using corrected HPTT01/HPPT01 suction superheat, with HPPT06 as the pressure immediately after the EEV and before the evaporator. There is no recorded compressor start/stop contact. A 0% HPCO01 command can mean stopped or operation at the 30 Hz minimum, so each event below is inferred from multiple process signals.

EEV shutdown logic The hardcoded shutdown sequence normally set HPEV01 to 0%. Because HPEV01 was later found to be percent closed, that recorded 0% corresponds to approximately 100% physical opening, not a closed EEV. Some shutdowns did not execute even this programmed transition. The report therefore records the observed signal action separately from the desired physical isolation.

Estimated start	HPPT01	HPPT06	First 2 min suction condition	Evidence / assessment
09:30:30	179 kPa(g)	223 kPa(g)	SH median -1.9 K	High confidence from command and high-side rise; wet start.
12:07:20	120	120	SH median -1.9 K	Probable 30 Hz start while HPCO01 remains 0%; command rises at 12:08. Wet.
12:37:46	234	611	SH median -1.9 K	High-confidence commanded start; wet.
12:55:52	304	611	SH median -2.6 K	High-confidence commanded start; wet.
~13:55-13:56	~80 first valid	~81 first valid	SH median -1.1 K	Stop/restart occurs within telemetry gap; already operating at probable 30 Hz before command rises. Wet.
14:05:50	414	417	SH median -1.2 K	Probable 30 Hz restart after hardware cutoff, inferred from high-side rise and suction pump-down. Wet.
14:51:50	77	115	SH median +9.4 K; min +8.1 K	Probable 30 Hz start; command rises at 14:54:28. Only clearly dry startup.

The previous command-only interpretation incorrectly treated the 13:22 and 13:49 rises as starts. They are now classified as speed increases from probable 30 Hz operation. Likewise, the command drops at 12:22, 13:15 and 13:37 are not accepted as physical stops without corroborating pressure and temperature behaviour.

Estimated stop	Key process evidence	HPPT06 / suction SH near stop	Classification
~11:35:32	After a command-zero/30 Hz pump-down, HPPT02 continues rising until low pressure is pulled down; HPEV01 changes to 0% at 11:35:36.	HPPT06 crosses 19.5 kPa at 11:35:20; SH rises to +23 K.	Good pressure/SH pump-down by configured logic; physical motor stop inferred, not recorded. Programmed EEV action is backwards physically.
~12:29:20	HPCO01 has been 0% since 12:22, but high-side behaviour and falling low-side pressure indicate continued 30 Hz pump-down. HPPT02 peaks then decays.	HPPT06 crosses 19.5 kPa; SH about +13 K.	Pressure criterion achieved; probable planned stop. HPEV01 stays near 5% recorded closed rather than executing the programmed 0% transition.
~12:48:45	HPPT02 immediately decays after command goes to 0%; HPEV01 stays near 4% recorded closed and HPPT06 remains at 611 kPa.	No pump-down; SH about -1.5 K.	Bad/possibly emergency or incorrectly sequenced stop; high confidence that compression ceased, but no run contact.
13:51-13:55 gap	HPPT06 is 32 kPa and SH +28 K at the last valid point; the next valid record shows HPEV01=0% and evidence of a restart.	Threshold crossing and exact stop/restart times are unobserved.	Probable pump-down and stop, but timing and sequence remain uncertain because telemetry is missing.
14:03:04	HPPT02 peaks at 2765 kPa then collapses while command stays 72%.	HPPT06 about 611 kPa; SH about -1.6 K.	Confirmed process signature of the emergency hardware high-pressure cutoff; no pump-down.
~14:10:40	Probable 30 Hz post-trip pump-down; HPEV01 changes from 95% to 0% at about 14:11 and HPPT02 decays.	HPPT06 crosses 19 kPa at 14:10:20; SH about +38 K.	Good pressure/SH pump-down by configured logic; physical stop inferred.
~16:40:05	Staged unload, high-side decay and pressure equalisation; HPEV01 moves toward 100% recorded closed.	HPPT06 about 222 kPa; SH approaches 0 K.	Orderly mechanical stop/cool-down but bad refrigerant-side pump-down; liquid margin inadequate.

Revised shutdown conclusion There are two strong pump-down signatures that reach the 20 kPa(g) HPPT06 target with positive suction superheat (about 11:35 and 14:10), plus a probable target-reaching stop near 12:29. The 12:49 and 16:40 stops do not pump down, and the 14:03 event is an emergency hardware cutoff. Exact motor stop times remain estimates because the start/stop contact and VSD actual-run status were not logged.

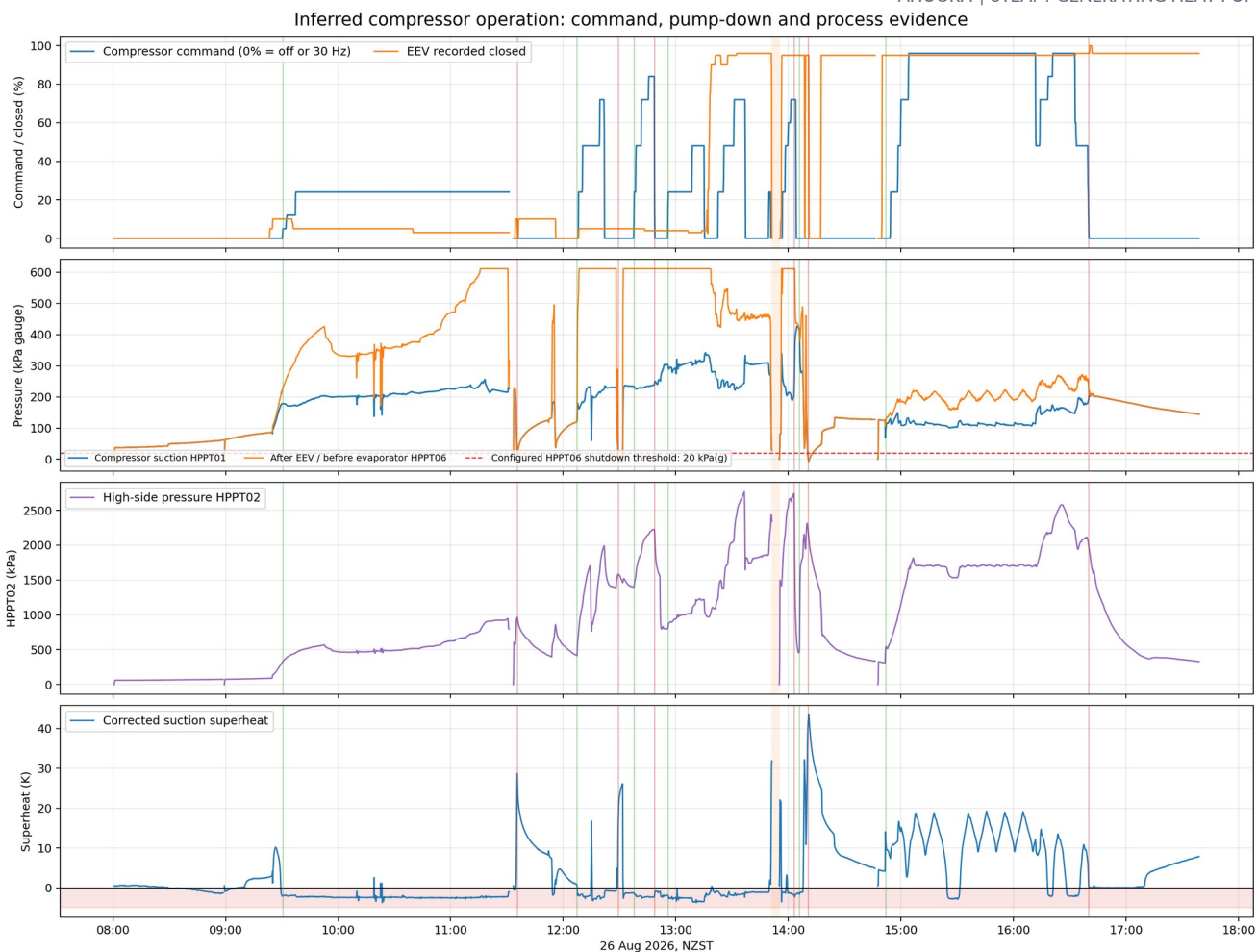


Figure 7. Inferred compressor operation. HPCO01=0% means either stopped or 30 Hz. Green/red lines are estimated starts/stops from combined evidence, not contacts; orange shading is the 13:51–13:55 telemetry gap. The dashed line is the 20 kPa(g) HPPT06 shutdown threshold.

Required shutdown record Log the intended sequence explicitly: isolate liquid feed using the corrected EEV direction and/or liquid-line solenoid, retain compressor operation within OEM limits, verify HPPT06 reaches 20 kPa gauge, verify HPPT01 is reduced and suction superheat positive, confirm the VSD/compressor stop contact, then retain water-side cooldown.

7. Steady-state assessment

A 10-minute candidate window was classified as locally steady when compressor command was constant, HPPT02 variability was below 1%, source-temperature standard deviation below 0.5 °C, and key temperature slopes below 0.2 °C/min. This identifies local operating plateaus, not necessarily safe or useful performance points.

Window / condition	Classification	Reason
09:59–10:19, ~24% compressor	Low-load quasi-steady	Commands, HPPT02 and source temperatures are locally stable, but suction remains wet/at saturation and no steam is produced. Not a valid performance point.
10:24–10:35 and ~11:20–11:30	Short candidate plateaus	Locally low rates of change, but the plant remains wet and/or slowly drifting; insufficient for performance claims.

Window / condition	Classification	Reason
15:05–16:11, 96% compressor, vent open	Cyclic steady state only	Steam remains at atmospheric boiling condition while source temperature and suction superheat repeat a ~9.7 min limit cycle. Time averages over whole cycles may be repeatable, but instantaneous plant state is not steady.
16:12–16:26, STGV61 closing	Transient pressurisation	Valve, high-side pressure and temperatures are moving; explicitly not steady state.
Overall	No full dry steam-production steady state	The only locally steady dry startup develops persistent source cycling and intermittent return to saturation.

7.1 Brief dynamic-response assessment

These are observed commissioning response times, not identified process-model constants. The compressor was manually ramped, STGV61 was closed in several steps, the source loop continued cycling, and actual compressor frequency was not logged; consequently a defensible first-order-plus-dead-time model cannot yet be fitted.

Observed transient	Approximate response time	Interpretation
Final startup from ~14:51:50	HPPT02 reached 63% of its rise to the first high-load plateau in 8.9 min and 90% in 12.3 min; STPT61 reached 100 °C in 14.0 min.	Overall warm-up is dominated by the deliberate compressor ramp and water/steam thermal inventory, rather than a single plant time constant.
STGV61 closure from 16:12:15	A clear HPPT02 rise appears in ~30 s. STPT61 reached 63% of its 24.5 K rise in 5.6 min; both STPT61 and HPPT02 reached ~90% of their observed rises in 11.3 min.	The steam-side pressure/temperature response is on a roughly 5–12 min timescale, but compressor-speed changes and gradual valve travel are confounding inputs.
14:03 hardware cutoff	HPPT02 completed 63% of its observed fall in ~32 s and 90% in ~72 s.	The refrigerant pressure state can change much faster than the steam-side thermal state; protection and event logging therefore need second-scale resolution.
Source-temperature cycling	Compressor-suction temperature follows SOTT31 within about 1 min; the dominant source cycle is ~9.7 min.	The source disturbance is fast enough to affect suction dryness well before the steam side settles.

Dynamic conclusion For preliminary controls work, treat the refrigerant-pressure response as a tens-of-seconds process, the source-to-suction path as about one minute, and the useful steam-side response as several minutes. Log actual VSD frequency and apply isolated test steps before estimating dead time, gain or PID tuning values.

8. Interesting patterns and provisional conclusions

- STGV61 closure is cleanly identifiable from 16:12:15 and causes coherent rises in steam temperature and high-side pressure; STTT61 fails to record the corresponding steam-pressure change.
- HPPT02 and HPPT03 agree extremely well during the final run (correlation 0.9989), supporting the high-side pressure trend.
- During pressurisation, the provisional absolute pressure ratio reaches 10.08 while HPTT02/HPPT02 calculates only 1.75 K median hot-gas superheat. Confirm the pressure basis and sensor locations before deciding whether this represents wet compression or a measurement-location problem.
- The final compressor probably starts at 30 Hz near 14:51:50, before HPCO01 rises above zero at 14:54:28. It remains the only clearly dry startup; all earlier inferred starts are wet.
- HPCO01=0% is not an off indication. Several former “stops” are reclassified as probable 30 Hz operation; the 13:22 and 13:49 command rises are speed increases rather than confident starts.
- Strong pump-down signatures reach the 20 kPa(g) HPPT06 target at about 11:35 and 14:10, with a probable third at 12:29. The 12:49 and 16:40 stops do not pump down; exact motor stop times are unverified.

- The shutdown logic normally wrote HPEV01=0%. Because the recorded signal is percent closed, that action physically opened rather than closed the EEV. Some shutdowns did not show the programmed transition at all.
- The 14:03 high-pressure cutoff is clearly distinct from a normal commanded stop: HPPT02 peaks at 2765 kPa and falls rapidly while HPCO01 remains at 72%, confirming the need for an actual VSD run/status signal.
- The auxiliary source-cooling cycle is normally highly repeatable, while the 18.25 min cycle is anomalous and coincides with the deepest source cooling and worst final-run liquid interval.
- WWTT41 and WWTT43 nearly duplicate SOTT31 dynamics. Confirm whether this reflects intentional hydraulic coupling, a common tank, colocated sensors or tag mapping.
- The only active automatic loop was flash-tank level; logging its PV, setpoint, output and mode is necessary to separate its effect from the source-side cycle.

9. Data dictionary corrections and exclusions

Item	Treatment for analysis
All temperatures	Convert raw counts using 0.1 °C/count.
All pressures	Engineering unit is kPa; record gauge/absolute reference per tag.
STPT61 data	Steam temperature; divide by 10. The tag name is misleading.
STTT61 data	Steam pressure in kPa; the tag name is misleading. Exclude until its lack of response during pressurisation is resolved.
HPCO01	Speed command relative to a 30 Hz minimum. A value of 0% means either stopped or running at 30 Hz; never classify compressor state from this topic alone.
HPEV01	Store both recorded % closed and derived % open = 100 – closed. Historical shutdown logic wrote 0%, which corresponds to fully open under the corrected direction.
HPEV01_Superheat	Retain for PLC diagnostic comparison only; do not call it compressor-inlet SH.
HPPT06	Post-expansion-valve / pre-evaporator pressure in kPa gauge. Use in pump-down review; configured stop threshold was 20 kPa(g). Quality-flag the repeated exact 611 kPa plateau.
Corrected suction SH	Compute from HPTT01 and HPPT01 using n-butane saturation and documented pressure reference.
STGV61	Exclude as valve feedback; no physical feedback signal exists.
STGV61O	Use only as requested valve output/command.
HPGD01 initial response	Annotate as expected detector startup sequence after extended power-off.
Common zero/sentinel values	Quality-flag; never silently treat as real process measurements.

10. Remaining questions

1. For each pressure transmitter, is the kPa value gauge or absolute, and what are its range and calibration date?
2. Why does STTT61 remain near 63–64 kPa during the known ~1 barg pressurisation: wrong PLC mapping, transmitter calibration, isolation/impulse-line problem, or wrong physical location?
3. Which PLC tags are flash-tank level PV, setpoint, controller output and auto/manual state?
4. Were the three final-run low/negative-superheat intervals seen physically as liquid carryover, compressor sound changes, accumulator level changes or frost/temperature changes?
5. Was the EEV held near 95% closed intentionally? Were any manual EEV adjustments made that are not represented in the command topic?
6. Which temperature is paired with HPPT06 in HPEV01_Superheat? P&ID V22 appears to label the evaporator-inlet pressure point HP-PT04, so reconcile the drawing and PLC tag revision.
7. Do operator observations or VSD history confirm the inferred physical stops near 11:35, 12:29, 12:49, within the 13:51–13:55 gap, 14:03, 14:10 and 16:40?

8. Was the probable 30 Hz restart/pump-down from about 14:06 to 14:10 intentional after the high-pressure hardware cutoff?
9. Which shutdowns were expected to execute the hardcoded HPEV01=0% action, and which were emergency/manual paths that intentionally bypassed it?
10. Does the repeated exact HPPT06 value of 611 kPa represent the transmitter upper range, a PLC clamp/sentinel or a real scaled value?
11. What manual actions correspond to each midday bump test, and were any stops protective trips rather than planned stops?
12. Does the auxiliary source-cooling heat pump have a run contact, demand, leaving-water setpoint, minimum-off timer or fault history that can be aligned with the 9–10 min cycles and abnormal 15:17–15:35 interval?
13. Why do WWTT41/WWTT43 and SOTT31 track within roughly 0.1–0.2 K: shared hydraulic inventory, expected exchanger behaviour, or duplicated/mislocated tags?
14. What OEM-approved suction-pressure target and minimum superheat should define a completed pump-down and safe restart for this compressor/accumulator arrangement?
15. Are HPTT02 and HPPT02 physically colocated on the compressor-discharge line upstream of the condenser, and are their response times suitable for calculating discharge superheat?
16. What pressure-ratio, discharge-temperature and discharge-superheat limits are approved by the compressor manufacturer for butane at these operating conditions?
17. What is the design refrigerant charge and expected receiver/accumulator inventory? Is 4 kg the calculated commissioning charge or an initial fill awaiting optimisation?
18. Where is electrical input power logged? Steam was vented and unmetered, so future capacity/COP work needs electrical power plus steam or condensate mass.

11. Recommended next run and analysis

- Build an authoritative tag dictionary: physical location, P&ID reference, engineering conversion, gauge/absolute basis, valid range and sentinel values.
- Create a timestamped operator log for bump tests, compressor commands, EEV adjustments, STGV61 actions, observed liquid carryover and steam observations.
- Recalculate all thermodynamic states from raw pressure/temperature pairs with explicit calibration and uncertainty bounds; retain raw and derived datasets separately.
- Loop-check HPTT02/HPPT02 and HPTT03/HPPT03, then add compressor-outlet and oil-separator temperatures plus receiver/accumulator level or inventory observations.
- Add explicit START_PREP, START_RAMP, DRY_RUN, PUMP_DOWN, COMPRESSOR_STOP and WATER_COOLDOWN states, with pump-down completion, dry-suction permissives and actual-state transitions recorded.
- Analyse suction dryness first: correlate corrected superheat with EEV inferred opening, compressor command, accumulator/flash-tank level, and the active level PID output.
- For steady-state performance, require a dry suction condition and hold all external cycles constant; otherwise report whole-cycle time averages as cyclic steady state.
- Future steam performance testing needs a condensate collection/weighing method or validated steam flow measurement; the commissioning steam was vented and unmetered.
- For the next run, log a compressor/VSD actual-run contact, start/stop request, run enable, actual frequency, current and power. Keep HPCO01 as command only.
- Log EEV requested opening, actual position if added, the corrected direction, shutdown-sequence state and liquid-line solenoid state.
- Log the complete high-pressure hardware trip chain: switch state, trip setpoint, VSD run enable, VSD actual frequency/status and reset event. HPCO01 alone does not show the 14:03 hardware stop.
- Log the auxiliary heat pump actual compressor state, demand, supply/return temperatures, setpoint, alarm/trip code and manual restarts alongside the source-loop signals.

Interpretation boundary This is a reconstruction from telemetry plus the commissioning team's clarifications. Start/stop contacts were not recorded, HPCO01=0% is ambiguous, and the estimated motor-state intervals rely on process response. Safety conclusions and performance claims still require confirmed compressor state, pressure scaling, calibration, flow/power measurements and operator observations.

Appendix. Provenance

Source: heat-pump.ahuratech.app, InfluxDB organisation lab, bucket data, mqtt_consumer/value grouped by topic. Data spans 26 August 2026 08:01–17:39 NZST. Overview values are 30 s means; detailed suction, HPPT06 and HPPT02 event analysis is 2 s. N-butane saturation properties were evaluated with CoolProp 7.2.0. Reference drawing: P&ID V22, Rev 22, 27 July 2026. Authentication credentials are intentionally excluded.

Mean aggregation can blur steps and extrema. Final trip/control analysis should return to unaggregated data around each event. Calculated steam saturation pressures use 101.325 kPa as atmospheric pressure and are contextual checks, not measurements.